



Bountiful alien oceans

A research indicates that some distinct alien oceans could be host to more life than Earth's oceans. <https://digit.in/sep19-09>



Inside Jupiter's storms

A radio wave image provided a glimpse into what happens beneath the ammonia clouds surrounding Jupiter. <https://digit.in/sep19-10>

HOW OLD IS THE MOON?

Despite being an old question, it is only recently that scientists are beginning to accurately determine the age of the Moon

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The Solar System was formed about 4.5 billion years ago, when the accretion disk around the young Sun began to coalesce into planets. What followed was a period of frequent cataclysms, with asteroids, comets and the small rocky planets moving closer to the Sun. The gas

giants and the ice giants moved outwards, away from the Sun during this period. The period is known as "giant planet migration", and scientists are in the process of investigating exactly why it occurred. Planets were still in the process of formation, in an embryonic state known as protoplanets. Collision events that occurred between all these celestial bodies were common. New data from the Juno mission suggests that a protoplanet which was ten times the size of the Earth struck Jupiter during this

period, and the gas giant is reeling to this day because of the impact.

Scientists don't yet know exactly how the Moon was formed. The leading hypothesis is that a Mars-shaped planet collided with the early Earth. This is the only collision event of that scale in the history of the Earth, if it took place. The ejected material, from both the Earth and the planet called Theia formed a cloud of material that orbited around the Earth. This material coalesced into the body we now call the Moon. The surface of the Moon was a magma ocean at this point, which cooled gradually over a period of a 100 million years. During this time, the heavier materials sank to the base of the lunar magma ocean, while the lighter materials rose to the